

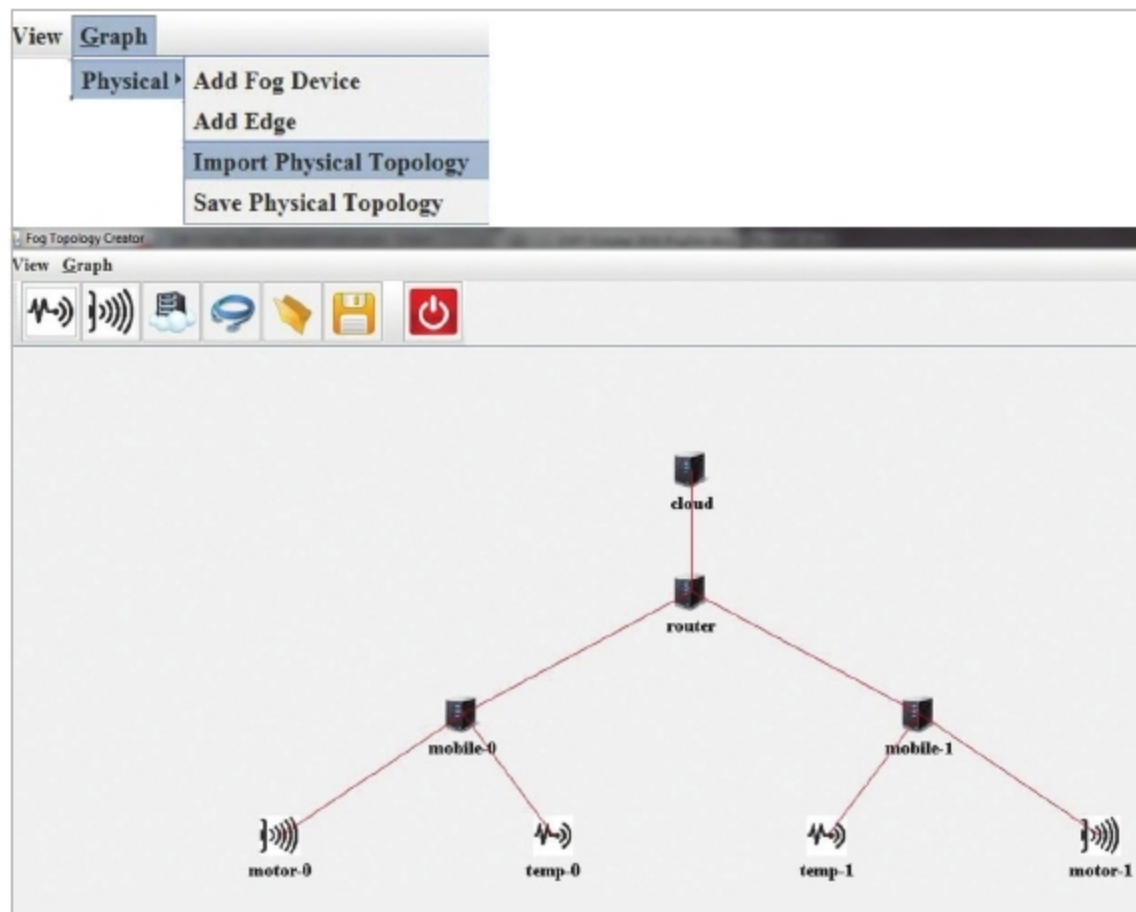


Figure 6: Importing the topology in the GUI of iFogSim

of intelligent surveillance in *org.fog.test.perfeval*. On executing it, various parameters like energy, cost, camera performance, etc, can be evaluated.

```
Starting DCNS...
Placement of operator object_detector on device d-0
successful.
Placement of operator object_tracker on device d-0
successful.
Creating user_interface on device cloud
Creating object_detector on device d-0
Creating object_tracker on device d-0
Creating motion_detector on device m-0-0
Creating motion_detector on device m-0-1
Creating motion_detector on device m-0-2
Creating motion_detector on device m-0-3
0.0 Submitted application dcns
=====
===== RESULTS =====
=====
EXECUTION TIME : 2583
=====
APPLICATION LOOP DELAYS
=====
[motion_detector, object_detector, object_tracker] --->
5.3571428571438195
[object_tracker, PTZ_CONTROL] ---> 3.110000000000363
=====
TUPLE CPU EXECUTION DELAY
=====
MOTION_VIDEO_STREAM ---> 2.957142857143481
DETECTED_OBJECT ---> 0.1111607142865978
```

```
OBJECT_LOCATION ---> 1.5285714285710128
CAMERA ---> 2.100000000000364
=====
cloud : Energy Consumed =
1.3338424452551037E7
proxy-server : Energy Consumed =
834332.9999999987
d-0 : Energy Consumed = 1048835.431000002
m-0-0 : Energy Consumed = 846301.761000042
m-0-1 : Energy Consumed = 846301.761000042
m-0-2 : Energy Consumed = 846301.761000042
m-0-3 : Energy Consumed = 846301.761000042
Cost of execution in cloud =
26120.742857167836
Total network usage = 11101.12
```

These are fully customisable libraries that can be improved further using new algorithms. These new algorithms can be programmed in the existing

libraries of iFogSim, so that the performance of the proposed or new algorithm can be analysed on fog based networks using iFogSim.

Scope for research and development

There are enormous challenges and areas of research in fog based networks including the following:

- Security, privacy and integrity
- Trust models and authentication
- Data aggregation and access control
- Improvements in the handover of networks
- Mobile fog computing
- Energy optimisation and resource management
- IoT microservices
- Quality of Experience (QoE)
- Smart grid architectures
- Offloading in fog networking
- Migration modelling
- Integration with machine learning and deep learning
- Virtualisation enabled fog
- Geospatial data analysis **END** 🐧

Reference

Harshit Gupta, Amir Vahid Dastjerdi, Soumya K. Ghosh, and Rajkumar Buyya: 'iFogSim: A Toolkit for Modelling and Simulation of Resource Management Techniques in Internet of Things, Edge and Fog Computing Environments'

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